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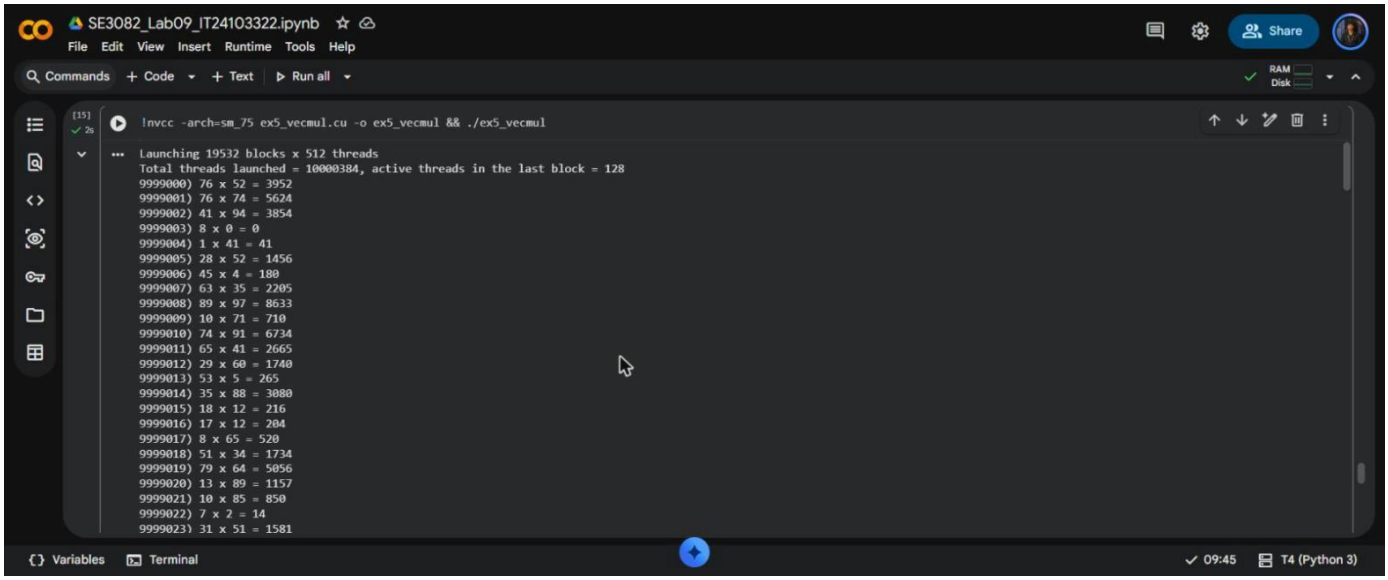
BSc (Hons) in Computer Science
Year 3

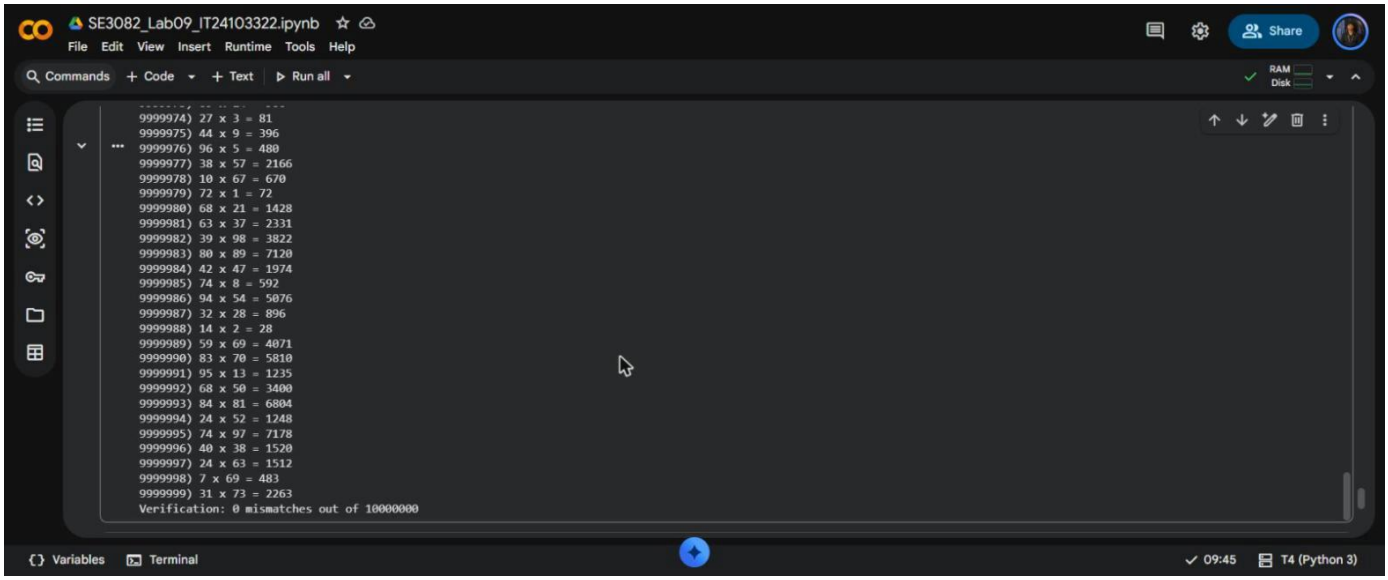
Lab Sheets

SE3082 – Parallel Computing **Semster 1, 2026**

IT24103322

Exercise 5





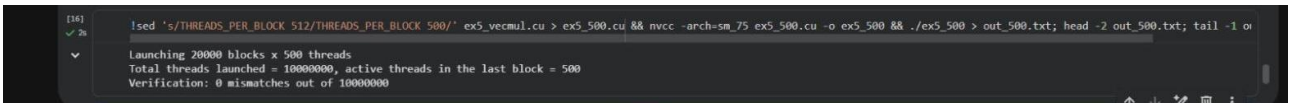
The kernel multiplies two vectors of 10,000,000 elements element by element, c[i] = a[i] * b[i], with 512 threads per block. Each thread finds its own element with the global index i = blockIdx.x * blockDim.x + threadIdx.x, and the line runs only if i < n. The number of blocks is rounded up, blocks = (N + 512 − 1) / 512

= 19532, so the threads cover every element. The last 1000 results (9999000 to 9999999) are printed, and all 10,000,000 results are checked on the CPU.

Item	Expected value	My result
Blocks launched	19532 blocks × 512 threads	Launching 19532 blocks x 512 threads
Total threads launched	10,000,384	10000384
Active threads in the last block	128	128
First of the last 1000 results	9999000) 76 x 52 = 3952	9999000) 76 x 52 = 3952
Last result	9999999) 31 x 73 = 2263	9999999) 31 x 73 = 2263
Verification	0 mismatches out of 10000000	0 mismatches out of 10000000

Every value matches the self-check. 19532 × 512 = 10,000,384 threads are launched, 384 more than the array, and only 128 threads of the last block have an element. Without the if (i < n) guard those 384 threads would read and write past the end of the arrays, which is undefined behaviour.

1. **Why is it a good idea to use a multiple of 32 for the threads per block? What happens to the total number of threads launched if you use 500 instead of 512?**



The GPU schedules threads in warps of 32. With a multiple of 32 every warp in a block is full. With 500 threads a block needs 16 warps (500 / 32 = 15.6), and the last warp has only 20 active threads and 12 idle lanes in every block.

With 500 threads per block the program launched 20000 blocks × 500 = 10,000,000 threads, exactly the size of the array, and still gave 0 mismatches. The number of threads goes down, but the hardware still runs 16 warps per block, which is 20000 × 512 = 10,240,000 lanes, compared with 10,000,384 for 512 threads per block. So 500 wastes more of the GPU even though it launches fewer threads.